Learning Theory

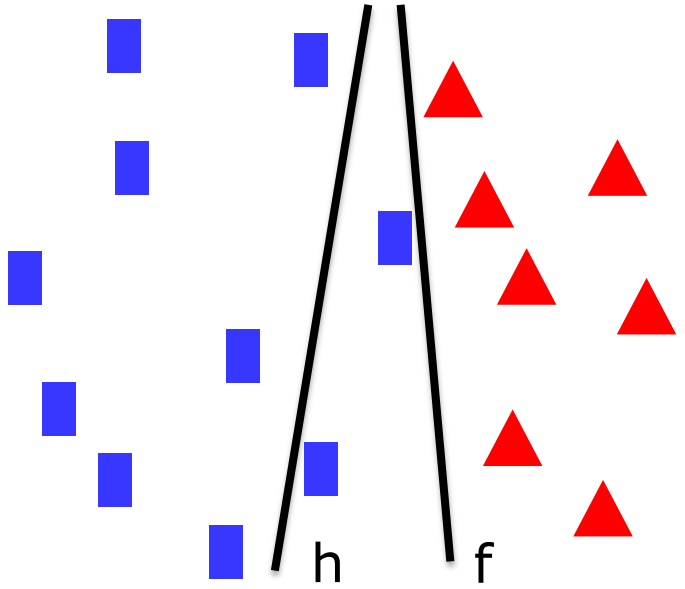
Finite Hypothesis Sets

Learning Theory

- Basic Questions:
  - Is learning even possible?
  - Can we give theoretical guarantees on quality of learned model?

Feasibility of Learning

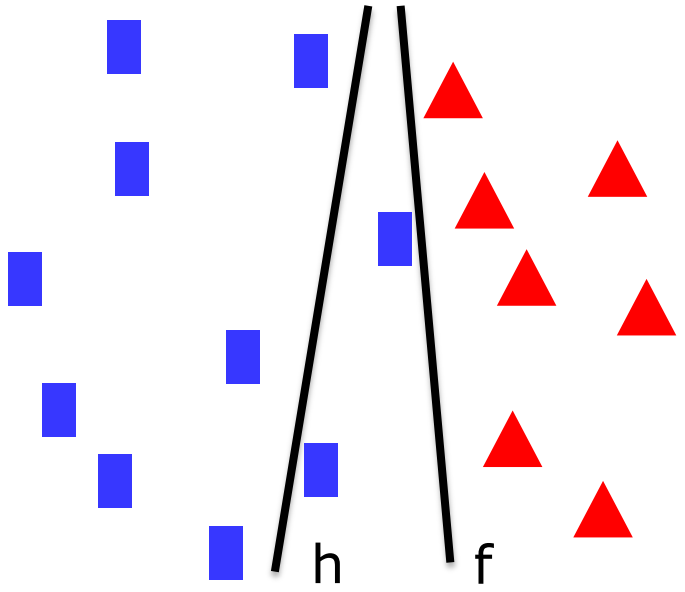
- Recall:
  - Trying to learn an unknown target function f:
    - Have data $(x_1,f(x_1))$, $(x_2,f(x_2))$, $(x_3,f(x_3))$,…



  - Goal: Find/learn hypothesis h ≈ f based on data

Feasibility of Learning

- What do we mean by h ≈ f ?
  - h has small error on new data!



- Is this possible in general?

Reasonable?

- Hope h ≈ f.
  - *Unknown* function f

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
f:	1	?	?	-1	?	-1	1	?

- Is it reasonable to predict values of ? based on values of $f(x_1)$, $f(x_4)$, $f(x_6)$, $f(x_7)$?

Reasonable?

- Hope h ≈ f.
  - *Unknown* function f
  - First question: Is it always the case that h ≈ f when $h(x_i)=f(x_i)$ on all training examples?

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
f:	1	?	?	-1	?	-1	1	?
h:	1	-1	1	-1	-1	-1	1	-1

- Is it reasonable to predict values of ? based on values of $f(x_1)$, $f(x_4)$, $f(x_6)$, $f(x_7)$?

Reasonable?

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f:	1	1	-1	-1	1	-1	1	1
h:	1	-1	1	-1	-1	-1	1	-1

NO!

- Is it reasonable to predict values of ? based on values of $f(x_1)$, $f(x_4)$, $f(x_6)$, $f(x_7)$?

What can we prove?

- Hope h ≈ f.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
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h:	1	-1	1	-1	-1	-1	1	-1

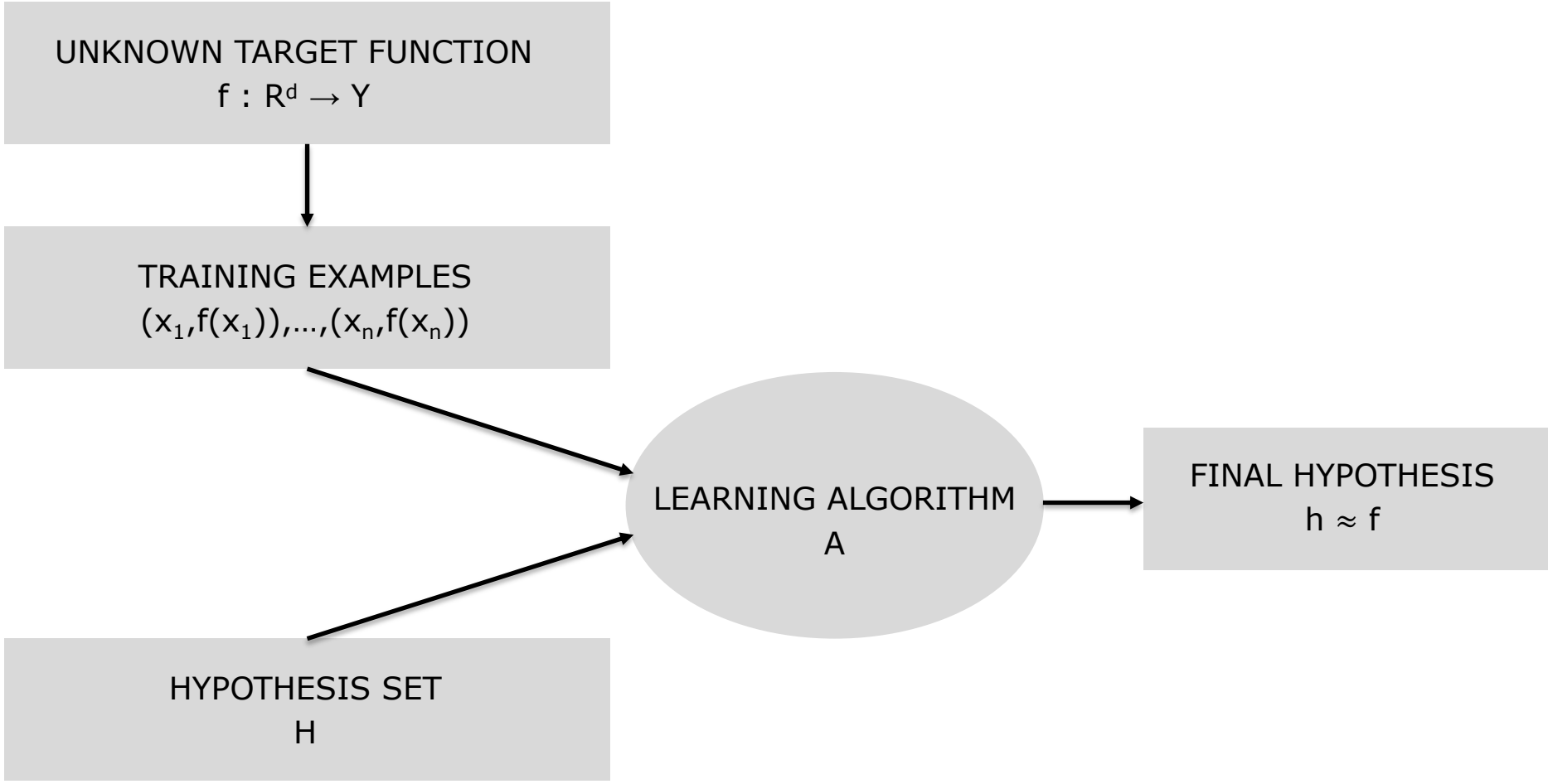
Is all hope gone?

Statistical Assumptions

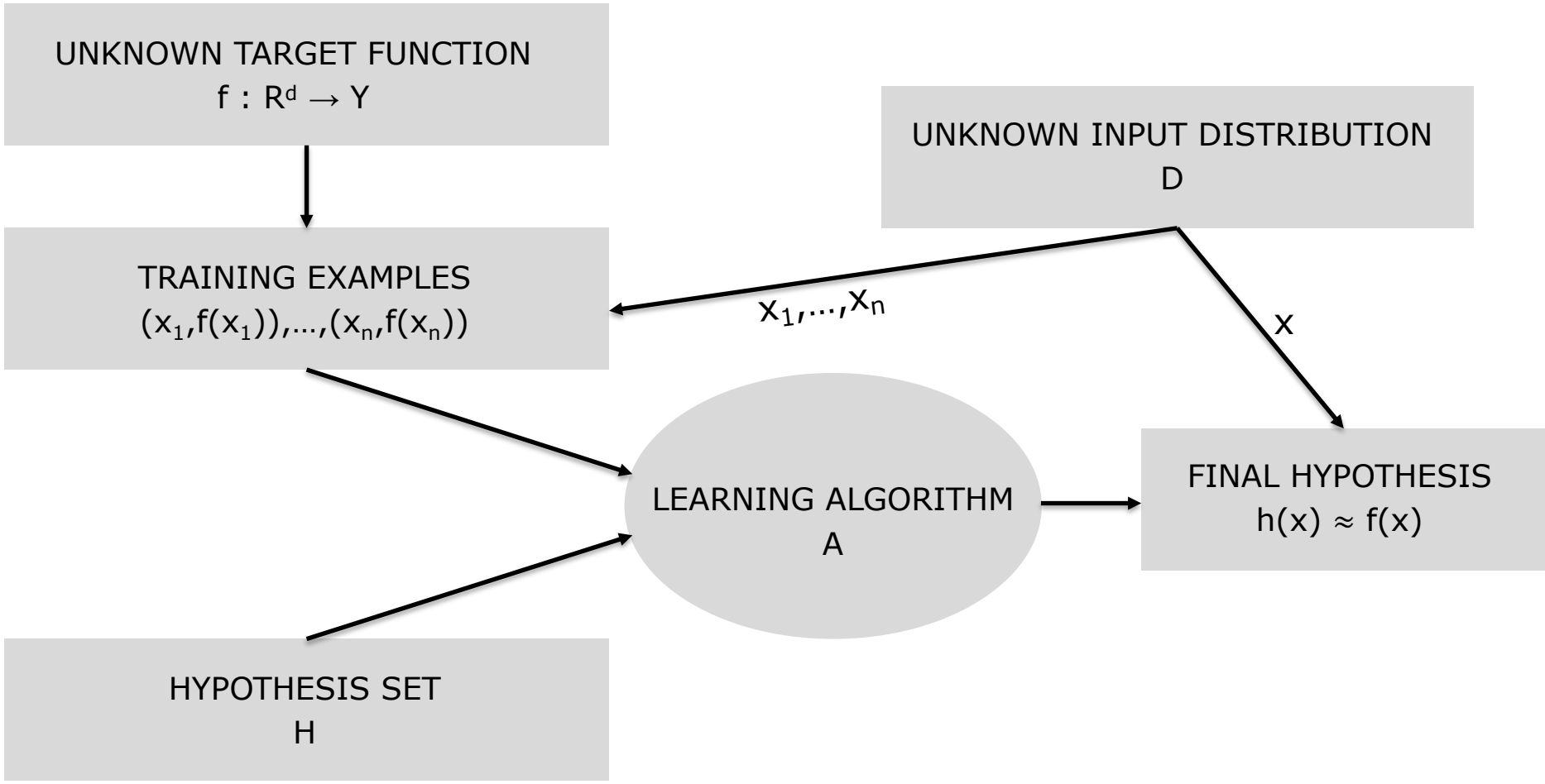
- Hope to prove h ≈ f.
- Strategy:
  - Make (reasonable) assumptions about how training data was obtained and how it relates to new data.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
f:	1	1	-1	-1	1	-1	1	1
h:	1	-1	1	-1	-1	-1	1	-1

Supervised Learning



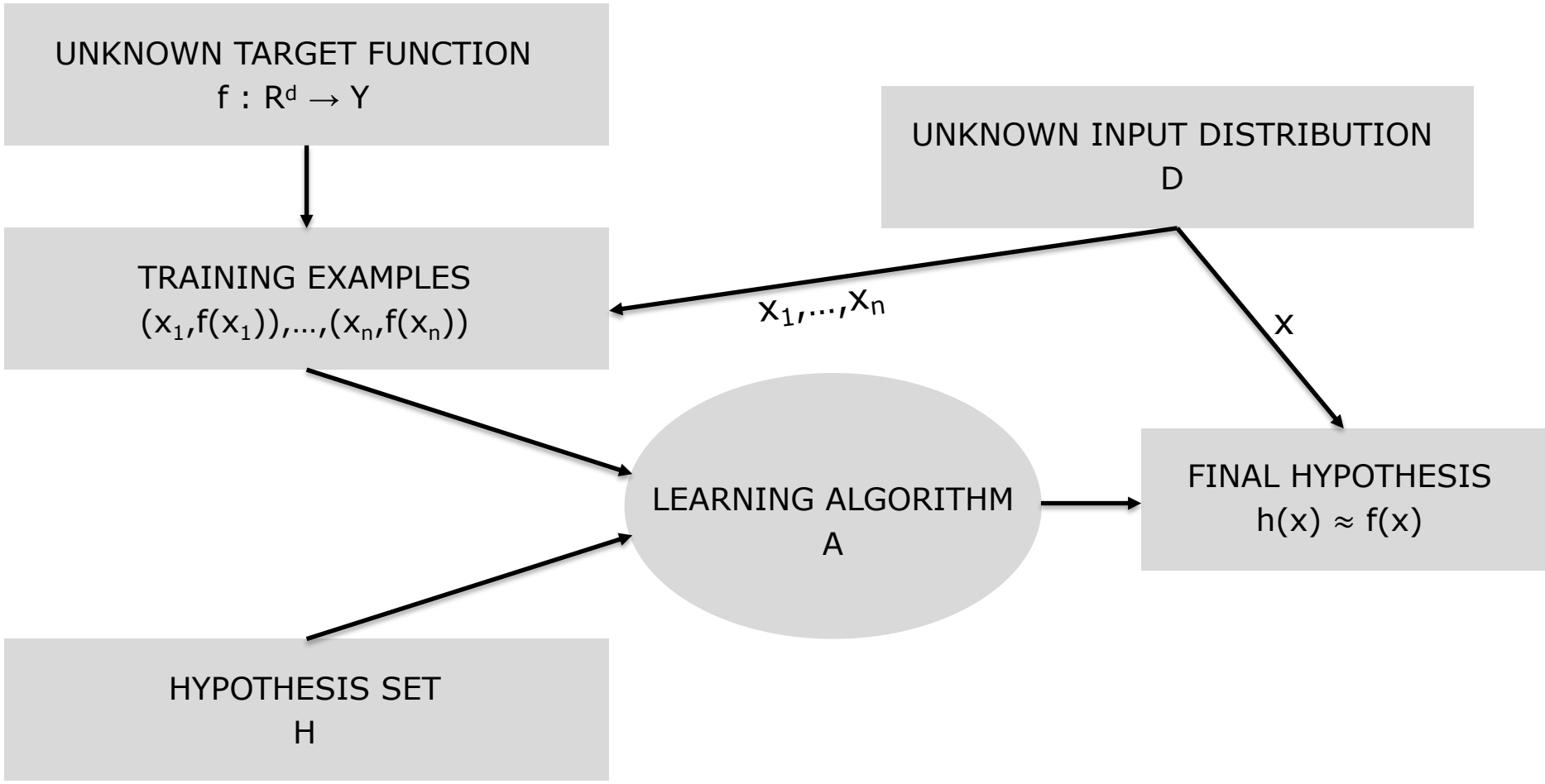
Supervised Learning



Statistical Assumptions

- Hope to prove $h \approx f$.
- Strategy:
  - Assume there is an *unknown* input distribution D on input feature vectors (Distribution on R^d).
  - Training data $(x_1,f(x_1)),\ldots,(x_n,f(x_n))$ obtained as follows:
    - Draw each x_i independently according to D.
  - Assume new unknown data follows same distribution:
    - A new example x to classify is drawn independently from D.
    - This is what we mean by new data similar to training data!
  - Goal: Prove that $h(x)=f(x)$ with good probability (over the distribution D).

Supervised Learning



Some Terminology

- We introduce two concepts:
  - In-sample error E_{in}:
    - The error on training data.
    - This is what we can try to minimize with a learning algorithm.
  - Out-of-sample error E_{out}:
    - The error on new unknown data.
    - This is what we actually want to minimize.

- Hope:
  - Can prove that $E_{in} \approx E_{out}$.
  - If so, then minimizing E_{in} also minimizes E_{out}.

In-Sample Error

- Data Generation:
  - Input domain R^d, output domain Y
  - Unknown distribution D over R^d
  - Unknown target $f : R^d \rightarrow Y$
  - Receive n samples $(x_1,f(x_1)),(x_2,f(x_2)),\ldots,(x_n,f(x_n))$.

This lecture

- 0-1 Loss from last lecture:

$$E_{in}(h) = \frac{1}{n}\sum_{i=1}^{n} 1_{h(x_i)\neq f(x_i)}$$

- General loss function L:

$$E_{in}(h) = \frac{1}{n}\sum_{i=1}^{n} L(h(x_i), f(x_i))$$

- Notation 1_q takes value 1 if q is true and 0 otherwise.

Out-of-Sample Error

- Data Generation:
  - Input domain R^d, output domain Y
  - Unknown distribution D over R^d
  - Unknown target $f : R^d \rightarrow Y$

This lecture

- 0-1 Loss from last lecture:

$$E_{out}(h) = Pr_D[h(x) \neq f(x)]$$

- General loss function L:

$$E_{out}(h) = \mathbb{E}_D[L(h(x),f(x))]$$

- Notation 1_q takes value 1 if q is true and 0 otherwise.

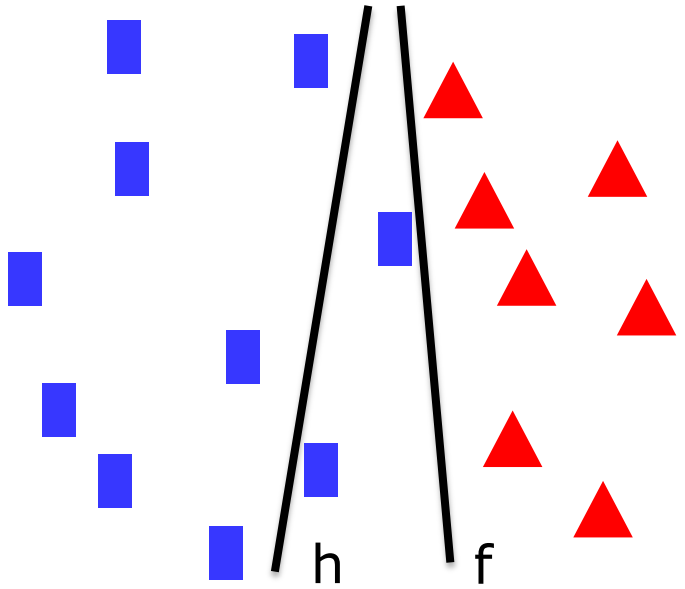
0-1 Loss:

$L(y',y) = 1_{y' \neq y}$

$\mathbb{E}_D[L(h(x),f(x))] = \mathbb{E}_D[1_{h(x) \neq f(x)}] = Pr_D[h(x) \neq f(x)]$.

Feasibility of Learning

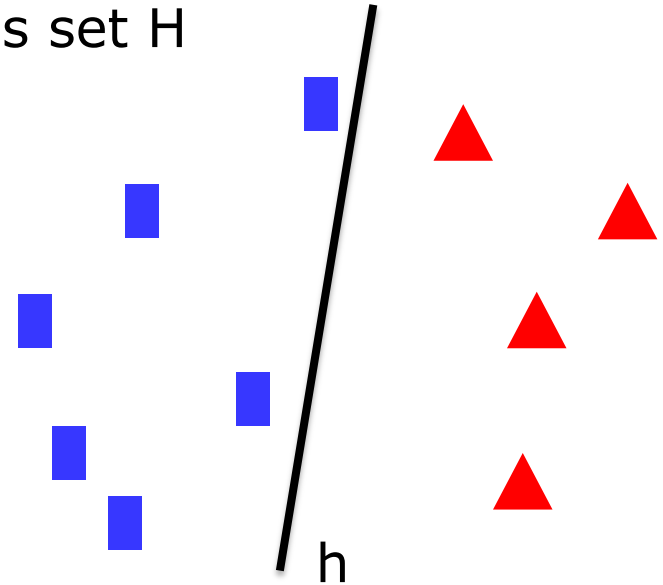
- What do we mean by h ≈ f ?
  - h has small error on new data!



- That is: The out-of-sample error $E_{out}(h) \approx 0$.
- Problem: Don't know f (and don't know D).

Learning

- Want: h with $E_{out}(h) \approx 0$.
- Have:
  - n samples $(x_1,f(x_1)),(x_2,f(x_2)),\ldots,(x_n,f(x_n))$
  - Hypothesis set H



H:
All hyperplanes/lines

- Strategy:
  - Find $h \in H$ with small in-sample error $E_{in}(h)$.
  - Hope $E_{in}(h) \approx E_{out}(h) \approx 0$

Reasonable?

- Hope $E_{in}(h) \approx E_{out}(h) \approx 0$.
  - *Unknown* function f

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
f:	1	?	?	-1	?	-1	1	?

- Is it reasonable to predict values of ? (out-of-sample) based on values of $f(x_1)$, $f(x_4)$, $f(x_6)$, $f(x_7)$ (in-sample)?

What can we prove?

- Hope $E_{in}(h) \approx E_{out}(h) \approx 0$.
- Relax requirement to $E_{in}(h) \approx E_{out}(h)$.

- First question: Is it always the case that $E_{in}(h) \approx E_{out}(h)$?

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
f:	1	1	-1	-1	1	-1	1	1
h:	1	-1	1	-1	-1	-1	1	-1

NO!

- Black is the sample, i.e. sample = $(x_1,1)$, $(x_4,-1)$, $(x_6,-1)$, $(x_7,1)$.
- $E_{in}(h) = 0$, $E_{out}(h) = ½$ (assuming uniform D)

What can we prove?

- Hope $E_{in}(h) \approx E_{out}(h) \approx 0$.
- Relax requirement to $E_{in}(h) \approx E_{out}(h)$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
f:	1	1	-1	-1	1	-1	1	1
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Is all hope gone?

What can we prove?

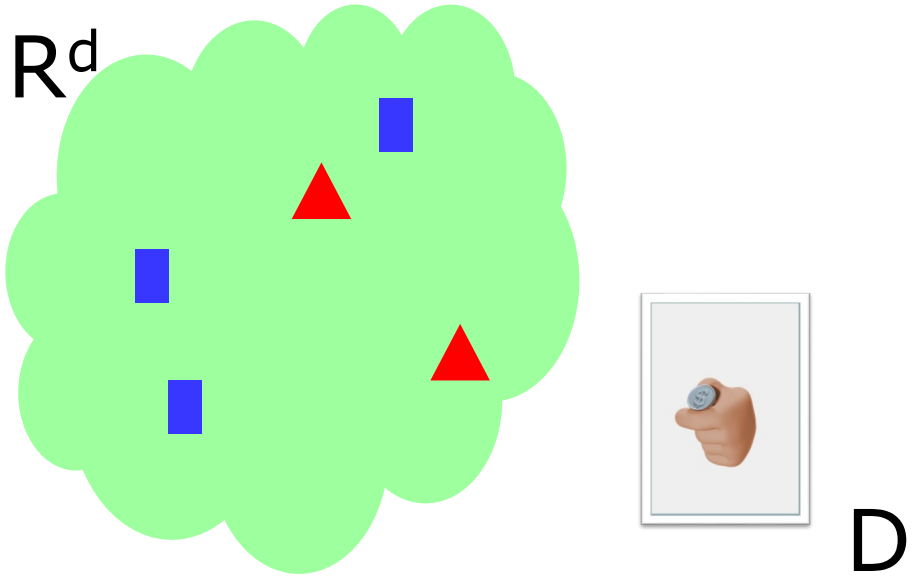
- Hope $E_{in}(h) \approx E_{out}(h) \approx 0$.
- Relax requirement to $E_{in}(h) \approx E_{out}(h)$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
f:	1	1	-1	-1	1	-1	1	1
h:	1	-1	1	-1	-1	-1	1	-1

- Key Observation:
  - If $E_{out}(h) = ½$, then unlikely to get a sample with $E_{in}(h) = 0$

What can we prove?

- Hope $E_{in}(h) \approx E_{out}(h) \approx 0$.
- Relax requirement to:
  - It is *probably* the case that $E_{in}(h) \approx E_{out}(h)$.



- With good probability *over the sample*, we have $E_{in}(h) \approx E_{out}(h)$.
- PAC Learning: Probably Approximately Correct.

One Hypothesis

- Setup:
  - One fixed hypothesis h. (Think H={h}).
  - Want to prove that $E_{in}(h) \approx E_{out}(h)$ with good probability over the sample.

Sample:

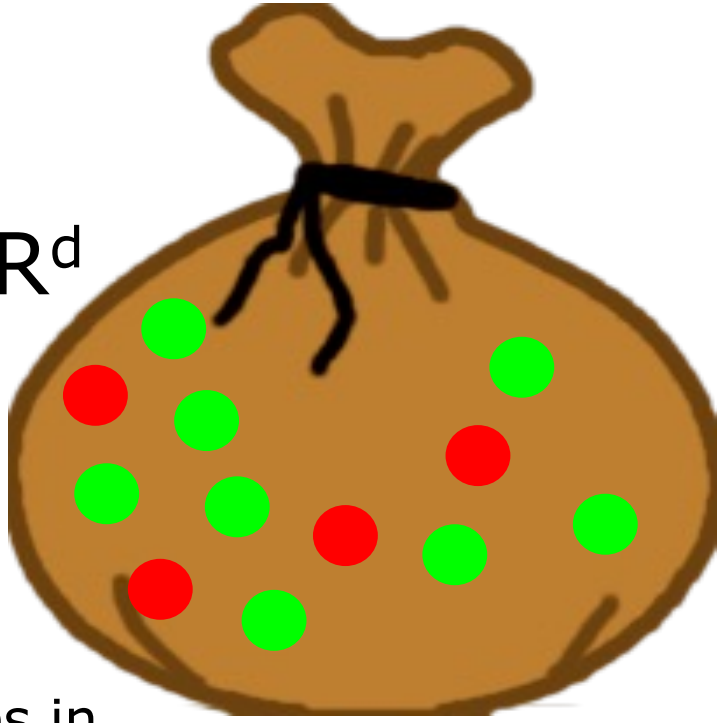
Pick n marbles (~D, with replacement)



$E_{in}(h)$:

Fraction of red marbles in sample.

R^d



Green(x): h(x)=f(x)

Red(x): h(x)≠f(x)

$E_{out}(h)$: Prob. of drawing red marble (~D).

One Hypothesis

$E_{in}(h)$: Fraction of red marbles in sample.

$\approx ?$

$E_{out}(h)$: Prob. of drawing red marble (~D).

Sample:

Pick n marbles (~D, with replacement)





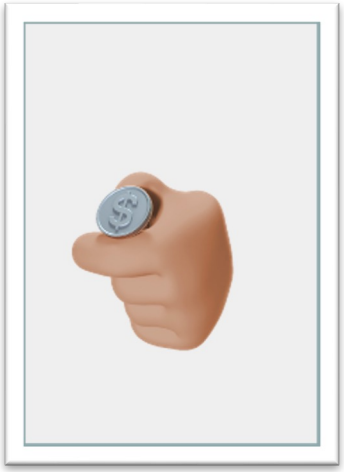
Different Analogy

$\approx ?$

$E_{in}(h)$: Fraction of red marbles in sample.

$E_{out}(h)$: Prob. of drawing red marble (~D).





Sample:

Toss biased coin n times, with prob. $E_{out}(h)$ of heads.

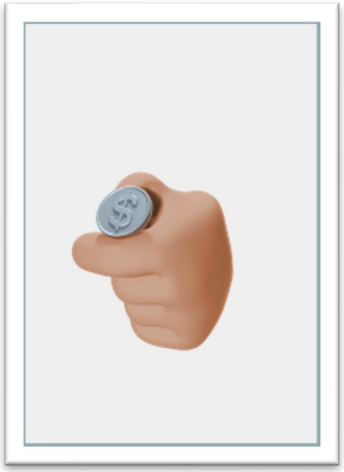
$E_{in}(h)$: Fraction of heads among tosses.

One Hypothesis

$E_{in}(h)$: Fraction of heads among n tosses.

$\overset{?}{\approx}$

$E_{out}(h)$: Prob. of heads

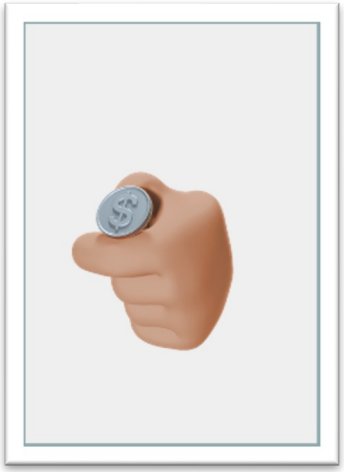


One Hypothesis

$E_{in}(h)$: Fraction of heads among n tosses.

$\approx ?$

$E_{out}(h)$: Prob. of heads



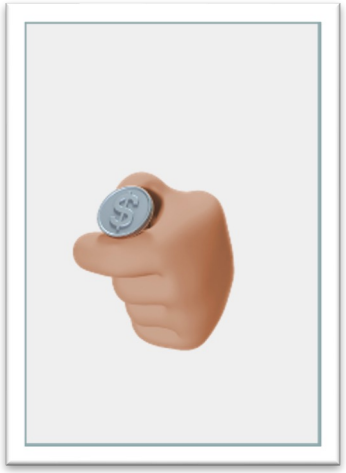
Binomial distribution:

$B(n, E_{out}(h))/n$

Well-studied!

Hoeffding's Inequality

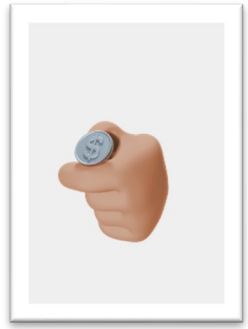
$E_{in}(h)$: Fraction of heads among n tosses. $\approx$ $E_{out}(h)$: Prob. of heads



Hoeffding:

$$\Pr[|E_{in}(h) - E_{out}(h)| > \varepsilon] \leq 2e^{-2\varepsilon^2 n}$$

Hoeffding's Inequality

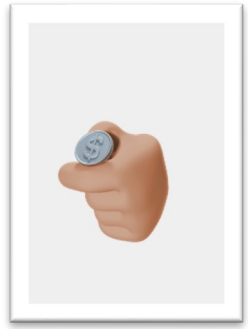


Hoeffding:

$$\Pr[|E_{in}(h) - E_{out}(h)| > \varepsilon] \leq 2e^{-2\varepsilon^2 n}$$

- Examples:
  - Sample size n=10
    - $\Pr[|E_{in}(h) - E_{out}(h)| > 0.1] \leq 2e^{-0.2} \approx 1.64$
    - $\Pr[|E_{in}(h) - E_{out}(h)| > 0.5] \leq 2e^{-5} \approx 0.01$
  - Sample size n=1'000
    - $\Pr[|E_{in}(h) - E_{out}(h)| > 0.1] \leq 2e^{-20} \approx 4 \cdot 10^{-9}$
    - $\Pr[|E_{in}(h) - E_{out}(h)| > 0.01] \leq 2e^{-0.2} \approx 1.64$
  - Sample size n=1'000'000
    - $\Pr[|E_{in}(h) - E_{out}(h)| > 0.001] \leq 2e^{-2} \approx 0.27$

What have we learned?



- If look at just one hypothesis h:
  - $\Pr[|E_{in}(h) - E_{out}(h)| > \varepsilon] \leq 2e^{-2\varepsilon^2 n}$

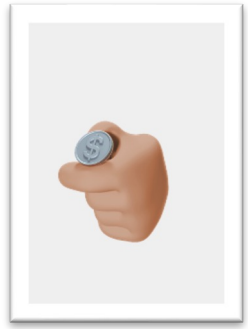
- Implications:
  - If lots of samples, then it is *highly likely* that $E_{in}(h) \approx E_{out}(h)$.
  - The randomness in *highly likely* is over the sample.

- In real life, we just have a data set.
- Interpretation is:
  - Assuming our data set was drawn i.i.d. from a distribution D, then it is likely that $E_{in}(h) \approx E_{out}(h)$.
  - The best theoretical statement we can prove.

Learning Theory

Finite Hypothesis Sets

What have we learned?

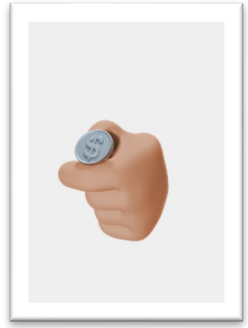


- If look at just one hypothesis h:
  - $\Pr[|E_{in}(h) - E_{out}(h)| > \varepsilon] \leq 2e^{-2\varepsilon^2 n}$

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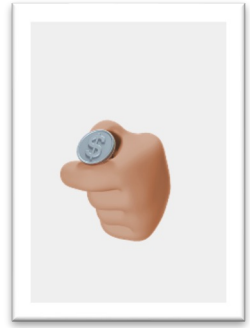
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- Interpretation is:
  - Assuming our data set was drawn i.i.d. from a distribution D, then it is likely that $E_{in}(h) \approx E_{out}(h)$.
  - The best theoretical statement we can prove.

Learning



- If look at just one hypothesis h:
  - $\Pr[|E_{in}(h) - E_{out}(h)| > \varepsilon] \leq 2e^{-2\varepsilon^2 n}$
- No learning!
  - When we run a learning algorithm, we search for an h* from a hypothesis set H.
  - We try to find an h* with small $E_{in}(h^*)$ and hope that $E_{in}(h^*) \approx E_{out}(h^*)$.
- Are we done?
  - Can we just plug in the h* returned by the learning algorithm and conclude $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2e^{-2\varepsilon^2 n}$?

Learning



- If look at just one hypothesis h:
  - $\Pr[|E_{in}(h) - E_{out}(h)| > \varepsilon] < 2e^{-2\varepsilon^2 n}$
- Are we done?
  - Can we just plug in the h* returned by the learning algorithm and conclude $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2e^{-2\varepsilon^2 n}$?

NO!

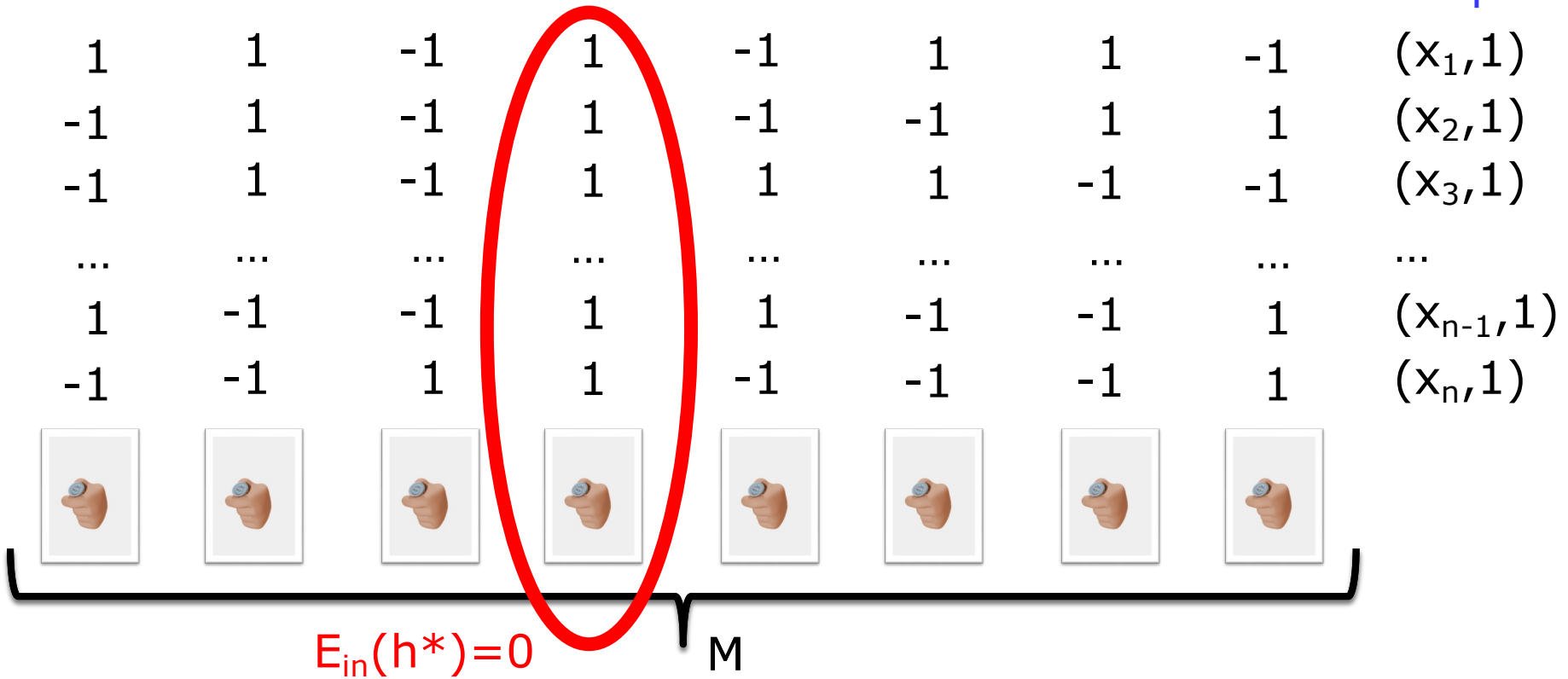
- Reason:
  - h* depends on the random choice of the sample.
  - Hoeffding bound doesn't hold.

Learning

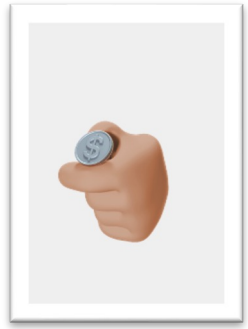
If M>2^n, then happens often

$\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2e^{-2\varepsilon^2 n}$ false

- If look at just one hypothesis h:
  - $\Pr[|E_{in}(h) - E_{out}(h)| > \varepsilon] \leq 2e^{-2\varepsilon^2 n}$
- Have H containing M hypotheses, all with $E_{out}(h)=1/2$



Multiple Hypotheses



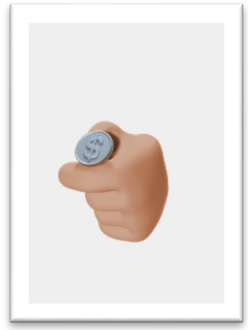
- If look at just one hypothesis h:
  - $\Pr[|E_{in}(h) - E_{out}(h)| > \varepsilon] \leq 2e^{-2\varepsilon^2 n}$

- How do we argue that $E_{in}(h^*) \approx E_{out}(h^*)$?
- We somehow need to use Hoeffding on hypotheses h that are chosen independently of the sample?

- Key Idea:
  - **If** $|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon$ **then**:
    $|E_{in}(h_1) - E_{out}(h_1)| > \varepsilon$ **or**
    $|E_{in}(h_2) - E_{out}(h_2)| > \varepsilon$ **or**
    …
    $|E_{in}(h_M) - E_{out}(h_M)| > \varepsilon$.

$H=\{h_1,h_2,\ldots,h_M\}$

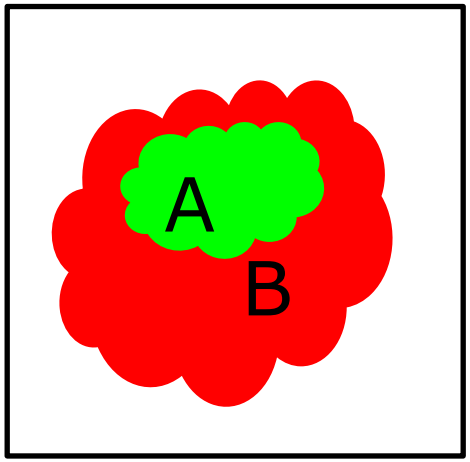
Multiple Hypotheses



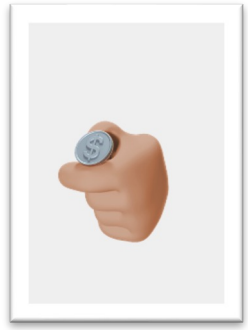
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    $|E_{in}(h_2) - E_{out}(h_2)| > \varepsilon$ **or**
    …
    $|E_{in}(h_M) - E_{out}(h_M)| > \varepsilon$.

- **If** A **then** B (A→B):
  - Pr[A] ≤ Pr[B]

Proof by picture



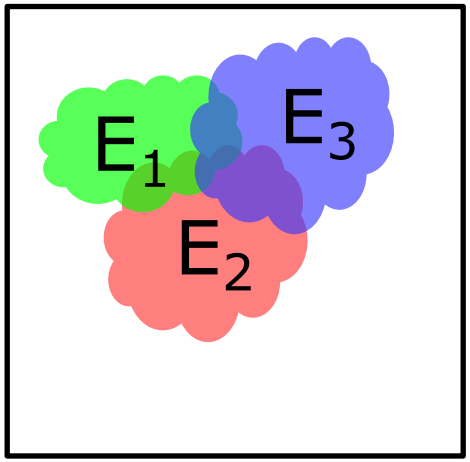
Multiple Hypotheses



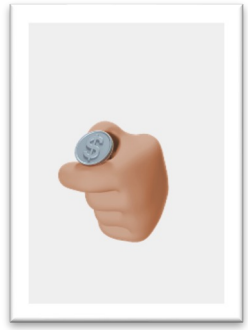
- $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq$
  $\Pr[|E_{in}(h_1) - E_{out}(h_1)| > \varepsilon$ **or**
  $|E_{in}(h_2) - E_{out}(h_2)| > \varepsilon$ **or**
  …
  $|E_{in}(h_M) - E_{out}(h_M)| > \varepsilon]$

- How do we handle all the or's?
  - Union bound!
  - $\Pr[E_1 \text{ or } E_2 \text{ or } \ldots \text{ or } E_M] \leq$
    $\Pr[E_1]+\Pr[E_2]+\ldots+\Pr[E_M]$.

Proof by picture



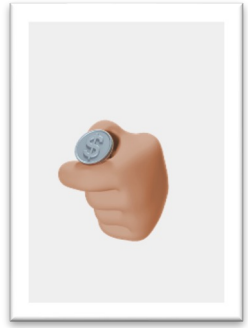
Multiple Hypotheses



- $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq$
  $\Pr[|E_{in}(h_1) - E_{out}(h_1)| > \varepsilon] +$
  $\Pr[|E_{in}(h_2) - E_{out}(h_2)| > \varepsilon] +$
  …
  $\Pr[|E_{in}(h_M) - E_{out}(h_M)| > \varepsilon]$

- Now the hypotheses in the events are chosen independently of the sample!
- We use Hoeffding:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$

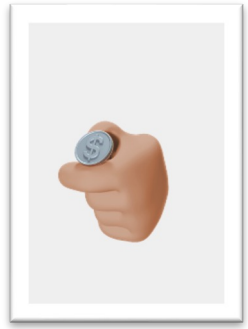
Multiple Hypotheses



- We proved:
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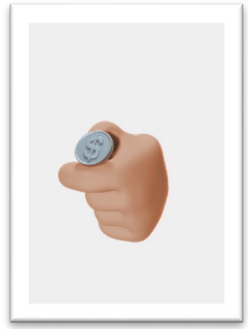
- Examples:
  - Sample size n=10, M=1'000:
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > 0.1] \leq 2'000e^{-0.2} \approx 1'637$
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > 0.5] \leq 2'000e^{-5} \approx 13$
  - Sample size n=1'000, M=1'000:
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > 0.1] \leq 2'000e^{-20} \approx 4 \cdot 10^{-6}$
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > 0.01] \leq 2'000e^{-0.2} \approx 1'637$
  - Sample size n=1'000'000, M=1'000:
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > 0.01] \leq 2'000e^{-200}$

Conclusions



- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$
- More samples n makes guarantees on closeness of $E_{in}(h^*)$ and $E_{out}(h^*)$ stronger.
- More complex hypothesis set H (larger M) makes guarantees on closeness of $E_{in}(h^*)$ and $E_{out}(h^*)$ weaker.
- That $E_{in}(h^*)$ and $E_{out}(h^*)$ are close ***does not*** guarantee small $E_{out}(h^*)$.
  - Could be the case that both $E_{out}(h^*)$ and $E_{in}(h^*)$ are large!
- Intuition: We hope for low $E_{in}(h^*)$ using small hypothesis set H.

Conclusions

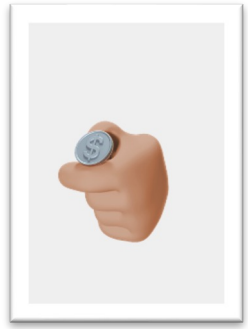


- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$
- Another Question:
  - Is it a good strategy to minimize $E_{in}(h^*)$ when running learning algorithm?

YES!

- Our proof in fact showed (via union bound) that ALL h in H simultaneously satisfy $|E_{in}(h) - E_{out}(h)| \leq \varepsilon$ with probability at least $1 - 2Me^{-2\varepsilon^2 n}$.

Conclusions



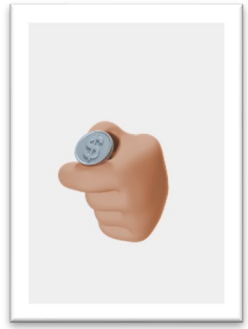
- Our proof in fact showed (via union bound) that ALL h in H simultaneously satisfy $|E_{in}(h) - E_{out}(h)| \leq \varepsilon$ with probability at least $1 - 2Me^{-2\varepsilon^2 n}$.

If $E_{in}(h^*)$ is smallest among all h in H, then for any other h' in H:

$$\begin{aligned} E_{out}(h^*) &= E_{out}(h^*) - E_{in}(h^*) + E_{in}(h^*) \\ &\leq \varepsilon + E_{in}(h^*) \\ &\leq \varepsilon + E_{in}(h') \\ &\leq \varepsilon + E_{in}(h') - E_{out}(h') + E_{out}(h') \\ &\leq 2\varepsilon + E_{out}(h') \end{aligned}$$

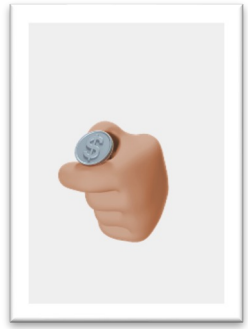
- Conclusion: Any other h' in H at most 2ε better E_{out}.

Conclusions



- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$
- Other formulation:
  - Assume we want with 95% confidence (over the sample), that $|E_{in}(h^*) - E_{out}(h^*)|<0.1$ and we use a hypothesis set with M=1'000 hypotheses, how many samples do we need?
- ε=0.1, M=1'000:
  - $2'000\, e^{-n/50} < 0.05$
  - $\leftrightarrow 40'000 < e^{\frac{n}{50}}$
  - $\leftrightarrow 50 \ln(40'000) < n$
  - $\leftarrow n > 530$

Issues

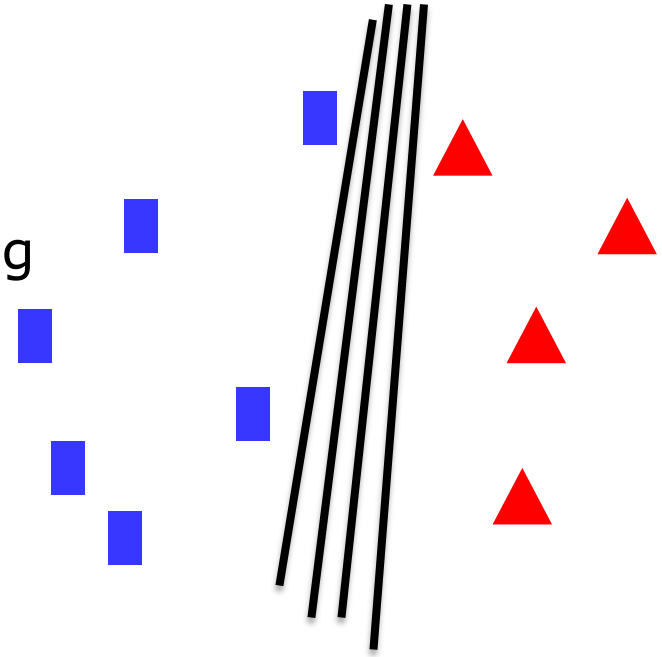


- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$
- Cannot handle infinite H (M=∞).

Finite Precision:

Use hyperplanes representable using d floats.

$M=2^{32d}$



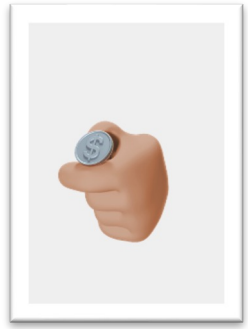
Non-trivial when:

If $n>(\varepsilon^{-2}/2)\ln(2^{32d+1})$

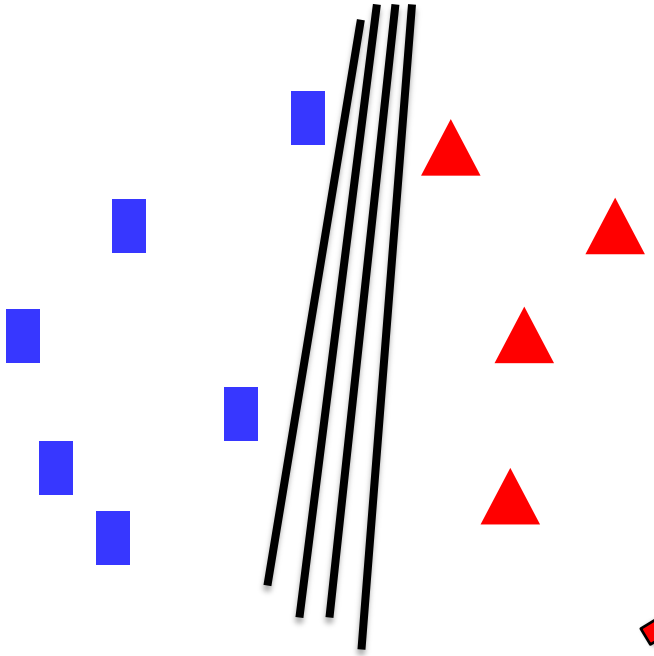
$\approx\varepsilon^{-2}$ 11d

Not too bad!

Issues

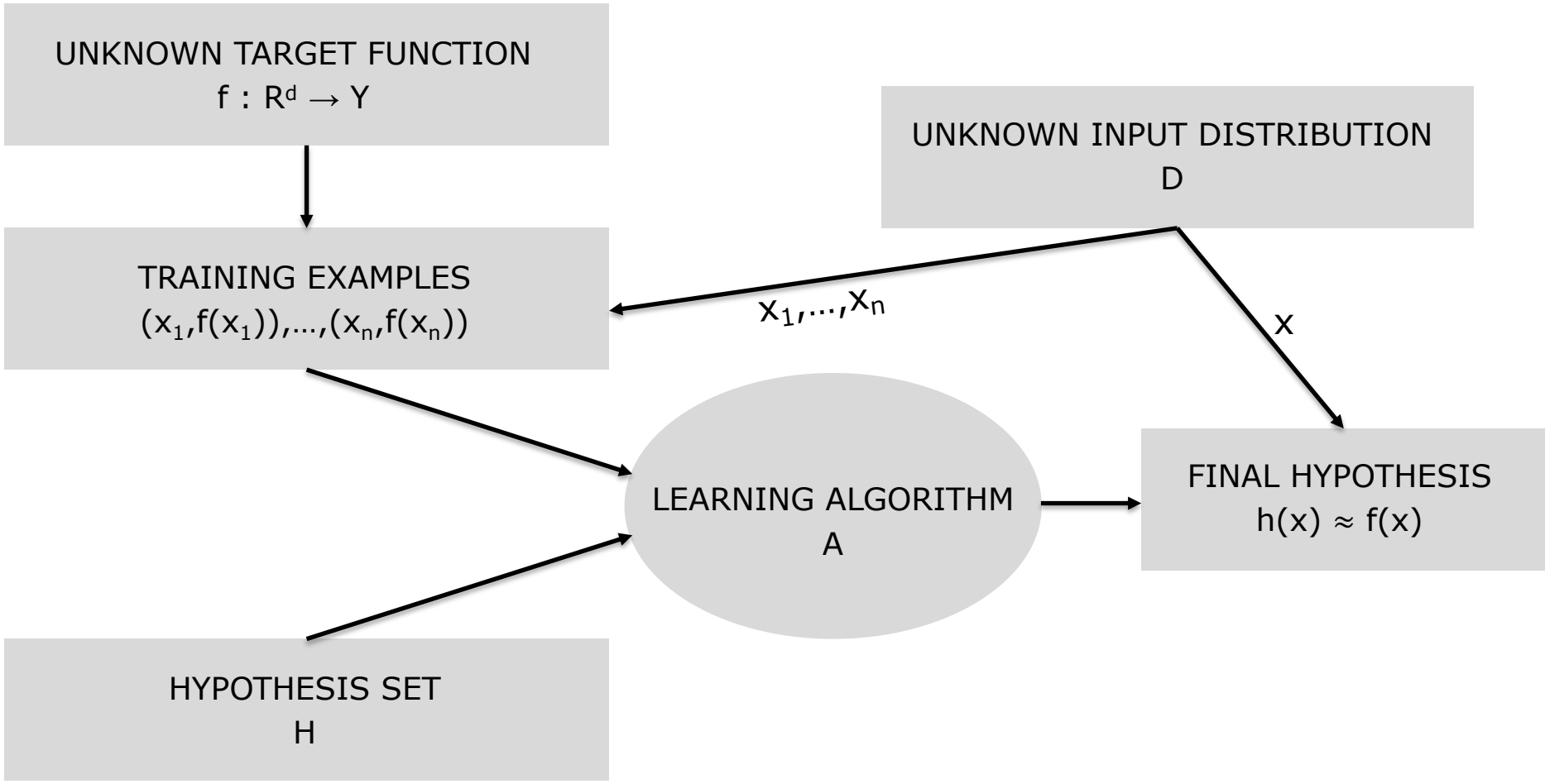


- We proved:
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- Cannot handle infinite H (M=∞).



Later: Infinite H

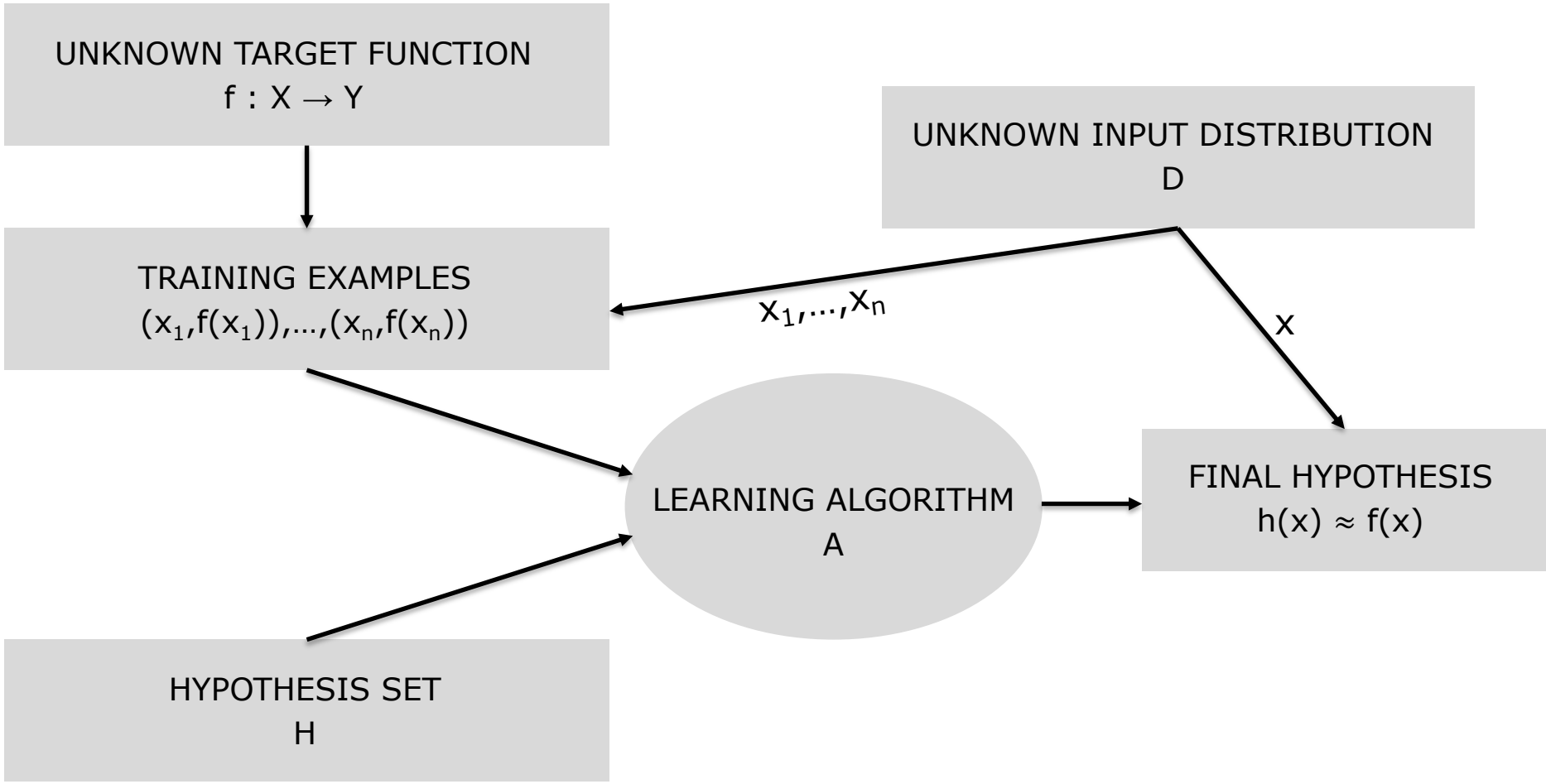
Some Extensions



Some Extensions

- Input Domain:
  - Input doesn't always have to be R^d.
    - It does for some learning algorithms, like perceptron.
    - But not for all.
  - For more generality, denote by X the input domain.
  - Could have $X=R^d$.
  - Could have $X=\{0,1\}^*$ (binary strings).
  - …

Some Extensions



Some Extensions

- Target Function:
  - We assumed unknown target function $f : X \rightarrow Y$.
    - Deterministic mapping from X to Y.
  - Issue: Not always the case that label/value completely determined from the available features.
    - Example: House price not completely determined from m^2 alone.
    - Example: Height of person not completely determined from age and gender.

- Model as follows:
  - For every x in X, there is a distribution on y, i.e. P(y | x).
  - Deterministic case corresponds to P(y | x) having all probability mass on a single element in Y.

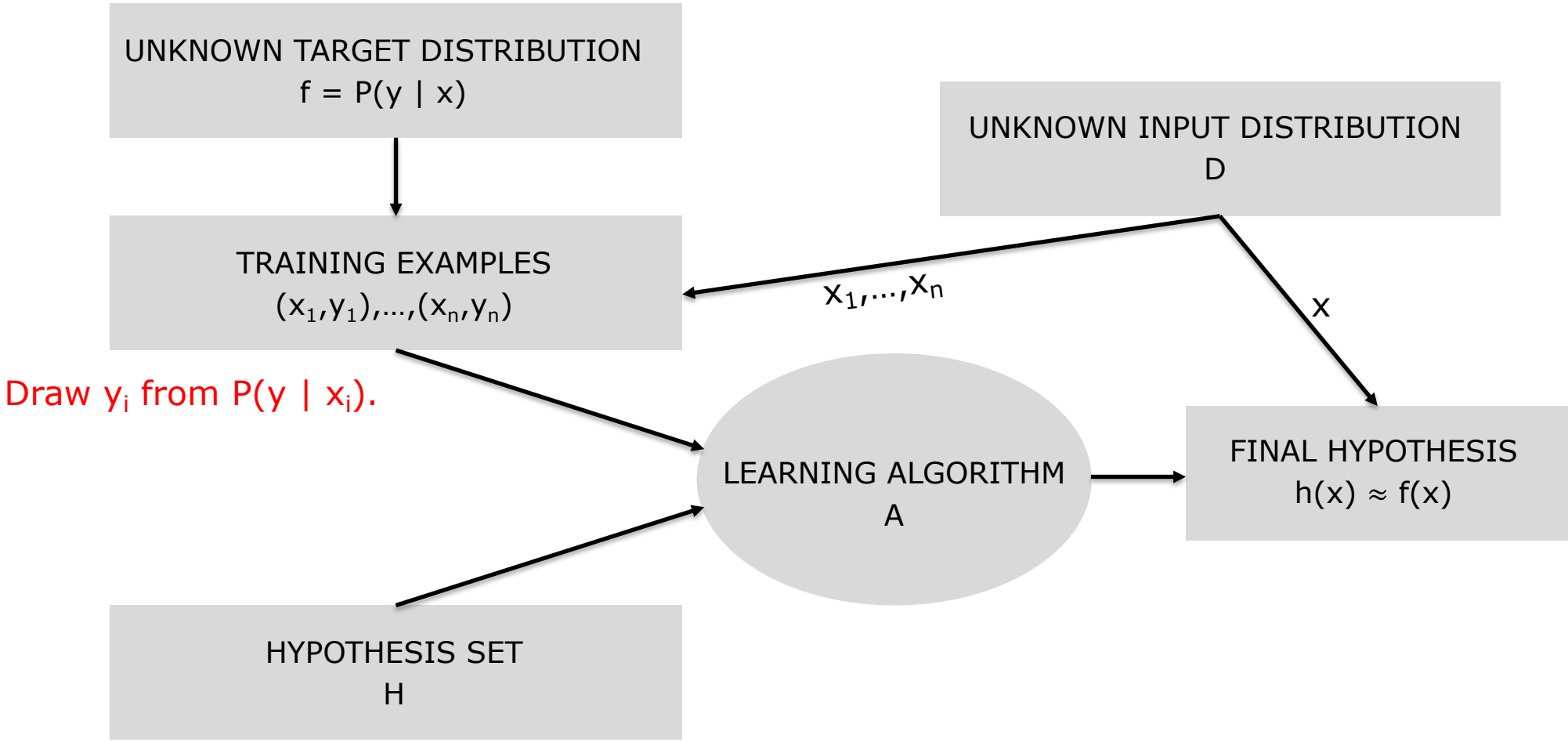
Some Extensions

- Target Function:
  - For every x in X, there is a distribution on y, i.e. P(y | x).
  - Deterministic case corresponds to P(y | x) having all probability mass on a single element in Y.

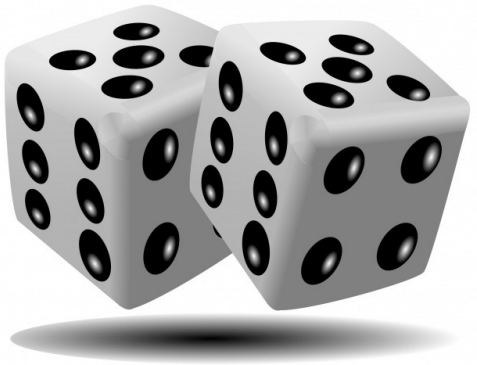
- Training data:
  - Obtained by drawing $x_1,\ldots,x_n$ from D. Then, for each x_i, generate y_i according to P(y | x_i).

- New unknown data:
  - Obtained by drawing x from D. Then, generate y according to P(y | x).
    - (x,y) with probability D(x)P(y | x).

Some Extensions



Some Extensions



- Example:
  - Experiment: Roll 2 dice.
  - Goal: Predict sum of dice.
  - Input features: Value of first die.

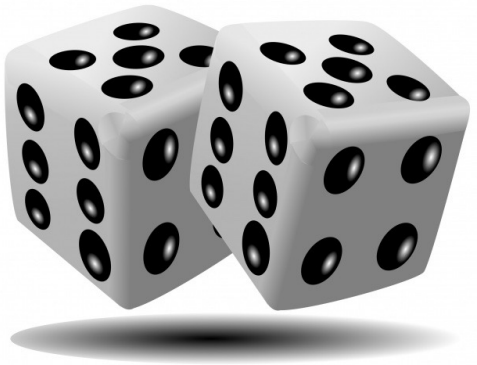
- X = {1,2,3,4,5,6}
- Y = {2,3,4,5,6,7,8,9,10,11,12}

P(y | 6)

2	3	4	5	6	7	8	9	10	11	12
0	0	0	0	0	1/6	1/6	1/6	1/6	1/6	1/6

Target not fully determined from available features.

Some Extensions



- Example:
  - Experiment: Roll 2 dice.
  - Goal: Predict sum of dice.
  - Input features: Value of first die.

- X = {1,2,3,4,5,6}
- Y = {2,3,4,5,6,7,8,9,10,11,12}

- Training Data Example:
  - (3, 7), (2, 3), (3, 9), (6, 8), (6, 7), …

  - Same x can have multiple labels.

In-Sample Error

- Data Generation:
  - Input domain X, output domain Y
  - Unknown distribution D over X
  - Unknown target distribution P(y | x)
  - Receive n samples $(x_1,y_1),(x_2,y_2),\ldots,(x_n,y_n)$.

x_i drawn from D

y_i drawn from P(y | x_i)

- 0-1 Loss from last lecture:

$$E_{in}(h) = \frac{1}{n}\sum_{i=1}^{n} 1_{h(x_i)\neq y_i}$$

- General loss function L:

$$E_{in}(h) = \frac{1}{n}\sum_{i=1}^{n} L(h(x_i), y_i)$$

Out-of-Sample Error

- Data Generation:
  - Input domain X, output domain Y
  - Unknown distribution D over X
  - Unknown target distribution P(y | x)

- 0-1 Loss from last lecture:

$$E_{out}(h) = Pr_D[h(x) \neq f(x)]$$ Deterministic target

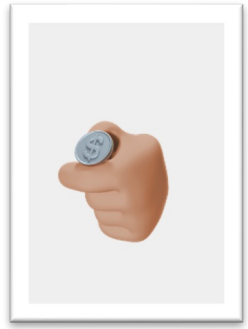
$$E_{out}(h) = Pr_{D,y}[h(x) \neq y]$$ Not deterministic target

- General loss function L:

$$E_{out}(h) = \mathbb{E}_D[L(h(x),f(x))]$$ Deterministic target

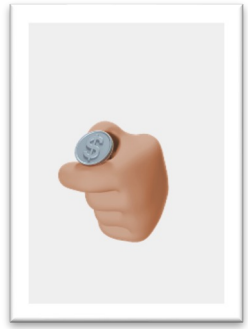
$$E_{out}(h) = \mathbb{E}_{D,y}[L(h(x),y)]$$ Not deterministic target

Conclusions



- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$
- Hoeffding also holds when target is not deterministic.
- For targets that are not deterministic, may be harder to get low $E_{in}(h^*)$.
- Example: P(y | x) is uniform random among -1 and +1 for all x.
  - $E_{out}(h^*)$ will always be ½.
  - $E_{in}(h^*)$ will not be much smaller (also according to Hoeffding).

Testing



- We proved:
  - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 2Me^{-2\varepsilon^2 n}$
- Gives us a way to bound $E_{out}(h^*)$.
  - We know $E_{in}(h^*)$ and can use formula.
- In practice:
  - If you need to estimate $E_{out}(h^*)$ instead do:
    - Set aside 20% of your data as a test set and don't use it for training.
    - Afterwards estimate $E_{out}(h^*)$ by computing accuracy of h^* on test set.
- We return to this and justify it later.